

Problem 3.1

Let $D = \overline{B}(a, r)$ be a closed Euclidean disk in $\mathbb{C}$ of radius $r > 0$ centered at $a \in \mathbb{C}$.

- (a) Suppose $z_1, \dots, z_n$ are points in D , and $\lambda_1, \dots, \lambda_n$ are real numbers in $[0, 1]$ with $\lambda_1 + \dots + \lambda_n = 1$. Show that $\lambda_1 z_1 + \dots + \lambda_n z_n \in D$.
- (b) Suppose $P(z) = \prod_{k=1}^n (z - z_k)$ is a complex polynomial whose zeros $z_1, \dots, z_n$ lie in D . Show that the zeros of P' also lie in D .

Solution. (a): Suppose $z_1, \dots, z_n \in D$ and $\lambda_1, \dots, \lambda_n \in [0, 1]$ where

$$\lambda_1 + \dots + \lambda_n = 1.$$

Then the claim follows from an immediate application of the triangle inequality:

$$\begin{aligned} \left| \sum_{k=1}^n \lambda_k z_k \right| &= \sum_{k=1}^n \lambda_k |z_k| \\ &\leq r \sum_{k=1}^n \lambda_k \\ &= r \end{aligned}$$

(b):

Problem 3.2

- (a) Let $S^1 := \{w \in \mathbb{C} : |w| = 1\}$. Show that $f : \mathbb{R} \rightarrow S^1, t \mapsto e^{it}$, is a covering map of $\mathbb{R}$ onto S^1 , i.e., for each $w_0 \in S^1$ there is an open neighborhood $U \subseteq S^1$ of w_0 such that $f^{-1}(U) = \bigcup_{k \in \mathbb{Z}} U_k$ for pairwise disjoint open sets $U_k \subseteq \mathbb{R}$ with $f|_{U_k}$ a homeomorphism of U_k onto U for each $k \in \mathbb{Z}$.
- (b) Show that $\exp : \mathbb{C} \rightarrow \mathbb{C}^* = \mathbb{C} \setminus \{0\}$ is a covering map.

Solution. Throughout, Arg denotes the principal branch of the argument, $\text{Arg}(z) \in (-\pi, \pi]$.

(a): $\forall w \in S^1, w = e^{it_k}$ where

$$t_k := \theta + 2\pi k, \quad \text{where } \theta := \text{Arg}(w) \in (-\pi, \pi], k \in \mathbb{Z}.$$

Take $U = B(w, \sqrt{2}) \cap S^1$. Take $U_k := \left(\theta + \frac{\pi(4k-1)}{2}, \theta + \frac{\pi(4k+1)}{2} \right) = \left(\theta + 2\pi k - \frac{\pi}{2}, \theta + 2\pi k + \frac{\pi}{2} \right)$. Note that $U_k \subset (\theta + \pi(2k-1), \theta + \pi(2k+1))$. Note that for distinct $a, b \in \mathbb{Z}$, $(\theta + \pi(2a-1), \theta + \pi(2a+1)) \cap (\theta + \pi(2b-1), \theta + \pi(2b+1)) = \emptyset$. Note that $f(U_k) = U$. Suppose that $\alpha, \beta \in U_k$ where $f(\alpha) = e^{i\alpha} = e^{i\beta} = f(\beta) \Rightarrow |\alpha - \beta| = 2\pi a$ where $a \in \mathbb{N}$. But the length of U_k , is π hence $a = 0$ and $\alpha = \beta$. Thus $f|_{U_k}$ is injective onto U hence $f|_{U_k}$ is a continuous bijection. It remains to show the inverse is continuous. Let $H := \{z \in \mathbb{C} : \text{Re}(z) > 0\}$. On H we have $\text{Arg}(z) \in (-\frac{\pi}{2}, \frac{\pi}{2})$ and $\sin(\text{Arg}(z)) = \text{Im} \frac{z}{|z|}$. Since $\arcsin : [-1, 1] \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$ inverts $\sin$ there,

$$\text{Arg}(z) = \arcsin\left(\text{Im} \frac{z}{|z|}\right) \quad (z \in H),$$

which is a composition of continuous functions, hence Arg is continuous on H . Define

$$g_k : U \rightarrow \mathbb{R}, \quad g_k(w) := \theta + 2\pi k + \text{Arg}(we^{-i\theta}).$$

If $w = e^{is} \in U$ with $|s - \theta| < \frac{\pi}{2}$, then $we^{-i\theta} = e^{i(s-\theta)} \in H$ with $s - \theta \in (-\frac{\pi}{2}, \frac{\pi}{2}) \subseteq (-\pi, \pi]$, so $\text{Arg}(we^{-i\theta}) = s - \theta$ and $g_k(w) = s + 2\pi k \in U_k$. Hence g_k is well defined, continuous, and $g_k(U) \subseteq U_k$. (We apply Arg to $we^{-i\theta} \in H$ rather than to w because Arg is discontinuous at -1 , and -1 may lie in U .) Moreover

$$f(g_k(w)) = e^{i\theta} e^{2\pi i k} e^{i \text{Arg}(we^{-i\theta})} = e^{i\theta} \cdot we^{-i\theta} = w,$$

i.e. $f|_{U_k} \circ g_k = \text{id}_U$. Since $f|_{U_k} : U_k \rightarrow U$ is a bijection, $g_k = (f|_{U_k})^{-1}$. Thus $f|_{U_k}$ is a continuous bijection with continuous inverse, i.e. a homeomorphism of U_k onto U .

(a): $\exp$ is surjective this is just a matter of checking using polar form. $\forall w \in \mathbb{C}^\times$, $w = re^{iy_k}$ where

$$y_k := \theta + 2\pi k, \theta \in (-\pi, \pi)$$

and $r > 0$. Let $x = \log(r)$. Let $U = \left\{ z \in \mathbb{C}^\times : \frac{z}{|z|} \in B\left(\frac{w}{|w|}, \sqrt{2}\right) \right\}$

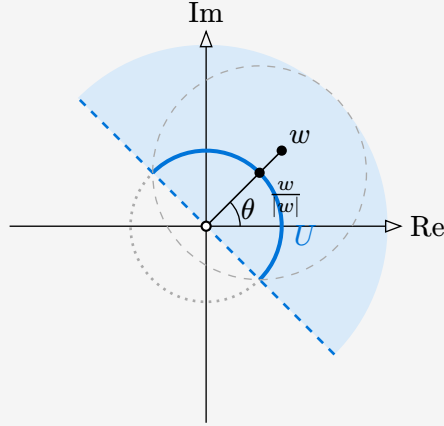


Figure 1: Let $w = 1 + i$. Then U is shown.

We have the following inverse image for U .

$$f^{-1}(U) := \bigsqcup_{k \in \mathbb{Z}} \mathbb{R} \times \left(\theta + 2\pi k - \frac{\pi}{2}, \theta + 2\pi k + \frac{\pi}{2} \right).$$

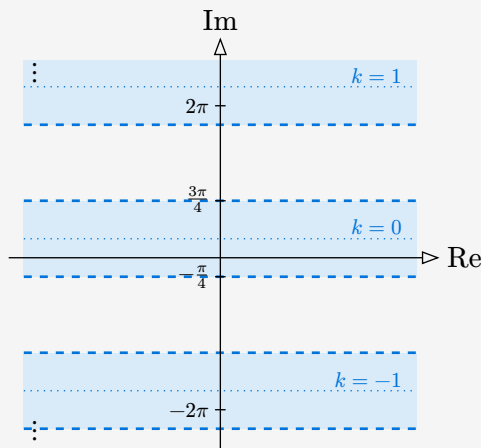


Figure 2: $\exp^{-1}(U)$ for $w = 1 + i$ ($\theta = \frac{\pi}{4}$): the disjoint open horizontal strips $\text{Im}(z) \in (\theta + 2\pi k - \frac{\pi}{2}, \theta + 2\pi k + \frac{\pi}{2})$, each of height π , shown for $k = -1, 0, 1$. The dotted centre line goes onto the ray through w .

Much like section 1, f is injective and surjective. The inverse $U \rightarrow \mathbb{R} \times U_k$ is given by $w \rightarrow \log(|w|) + ig_{k(z)}$ everything is continuous and behaves nicely

Problem 3.3

Let $P(z) = a_0 + a_1z + \dots$ and $Q(z) = b_0 + b_1z + \dots$ be polynomials with complex coefficients. Show that if $P(z) = Q(z)$ for all z , then $a_0 = b_0$, $a_1 = b_1$, etc.

Solution. $P(x) - Q(x) = 0$ everywhere. Then we see that $(P - Q)' = 0 \Rightarrow P - Q$ is a constant and hence 0. Thus

$$(P - Q)(x) = \sum_{k=0}^n (a_k - b_k)x^k = 0.$$

We know that $(P - Q)(0) = 0 \Rightarrow a_0 - b_0 = 0$. Then keep taking successive derivatives hence $P^{(k)}(0) = k!(a_k - b_k) = 0 \Rightarrow a_k = b_k$.

Problem 3.4

Let $U \subseteq \mathbb{C}$ be open and $z_0 \in U$. Suppose $f : U \rightarrow \mathbb{C}$ is differentiable at z_0 and $f(z_0) \neq 0$. Prove directly from the definitions that there is a neighborhood V of z_0 such that $g(z) = \frac{1}{f(z)}$ is defined on V , that g is differentiable at z_0 , and that

$$g'(z_0) = -\frac{f'(z_0)}{f(z_0)^2}.$$

Hint: do not apply the quotient rule; establish this special case directly.

Solution. Note that g is continuous at z_0 . $\forall \varepsilon > 0$, let $\delta_1 > 0$ such that

$$|z - z_0| < \delta_1 \Rightarrow \left| \frac{1}{f(z_0)} - \frac{1}{f(z)} \right| < \frac{\varepsilon |f(z_0)|}{2(|f'(z_0)| + 1)}.$$

Let $\delta_2 > 0$ such that

$$|z - z_0| < \delta_2 \Rightarrow |f(z) - f(z_0)| < \frac{|f(z_0)|}{2} \Rightarrow |f(z)| \geq \frac{|f(z_0)|}{2}$$

Let $\delta_3 > 0$ such that

$$|z - z_0| < \delta_3 \Rightarrow \left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| < \frac{\varepsilon |f(z_0)|^2}{4}.$$

Let $\delta = \min(\delta_1, \delta_2, \delta_3)$. Thus if $|z - z_0| < \delta$ then all three bounds are correct. Thus

$$\begin{aligned} \left| \frac{g(z) - g(z_0)}{z - z_0} + \frac{f'(z_0)}{(f(z_0))^2} \right| &= \left| \frac{f(z_0) - f(z)}{f(z)f(z_0)(z - z_0)} + \frac{f'(z_0)}{(f(z_0))^2} \right| \\ &= \frac{1}{|f(z_0)|} \left| \frac{f'(z_0)}{f(z_0)} - \frac{f(z) - f(z_0)}{f(z)(z - z_0)} \right| \\ &\leq \frac{|f'(z_0)|}{|f(z_0)|} \left| \frac{1}{f(z_0)} - \frac{1}{f(z)} \right| + \frac{1}{|f(z_0)||f(z)|} \left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| \\ &< \frac{|f'(z_0)|}{|f(z_0)|} \cdot \frac{\varepsilon |f(z_0)|}{2(|f'(z_0)| + 1)} + \frac{2}{|f(z_0)|^2} \cdot \frac{\varepsilon |f(z_0)|^2}{4} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \end{aligned}$$

where in the third line we split $\frac{f'(z_0)}{f(z_0)} - \frac{f(z) - f(z_0)}{f(z)(z - z_0)} = f'(z_0) \left(\frac{1}{f(z_0)} - \frac{1}{f(z)} \right) + \frac{1}{f(z)} \left(f'(z_0) - \frac{f(z) - f(z_0)}{z - z_0} \right)$, and in the fourth we used $|f(z)| \geq \frac{|f(z_0)|}{2}$ from δ_2 . Since $\varepsilon > 0$ was arbitrary, g is differentiable at z_0 with $g'(z_0) = -\frac{f'(z_0)}{(f(z_0))^2}$. Finally, δ_2 does not depend on ε , so $V := B(z_0, \delta_2) \cap U$ is a neighborhood of z_0 on which $|f| \geq \frac{|f(z_0)|}{2} > 0$, hence $g = \frac{1}{f}$ is defined on V .

Problem 3.5

Let $U \subseteq \mathbb{C}$ be a region and $u : U \rightarrow \mathbb{R}$ harmonic. Suppose v and w are harmonic conjugates of u on U . Show that $v - w$ is constant on U .

Solution.

Problem 3.6

Let $U \subseteq \mathbb{C}$ be a region and $f \in \mathcal{H}(U)$. Using the Cauchy–Riemann equations, show:

- (a) if $\operatorname{Re}(f)$ or $\operatorname{Im}(f)$ is constant on U , then f is constant.
- (b) if $|f|$ is constant on U , then f is constant.

Solution.

Problem 3.7

Let $U, V \subseteq \mathbb{C}$ be open, $h : V \rightarrow \mathbb{R}$ harmonic, and $f : U \rightarrow V$ holomorphic. Show that $g := h \circ f : U \rightarrow \mathbb{R}$ is harmonic. *Hint:* you may disregard smoothness issues. To show $\Delta g = 0$, use Wirtinger derivatives throughout.

Solution.

Problem 3.8

Let $U \subseteq \mathbb{C}$ be open. A C^2 function $s : U \rightarrow \mathbb{R}$ is *subharmonic* if $\Delta s \geq 0$ on U . Show that if $f : U \rightarrow \mathbb{C}$ is holomorphic, then $s(z) = \log(1 + |f(z)|^2)$ is subharmonic on U .

Solution.

Problem 3.9

Let u be a C^2 function on an open subset of $\mathbb{R}^2$, viewed as $u(x, y)$ and, via $x = r \cos \varphi$, $y = r \sin \varphi$, as a function of (r, φ) . Subscripts denote partial derivatives.

- (a) Show that $u_r = \cos(\varphi)u_x + \sin(\varphi)u_y$ and $u_\varphi = -r \sin(\varphi)u_x + r \cos(\varphi)u_y$.
- (b) Find similar expressions for u_{rr} and $u_{\varphi\varphi}$.
- (c) Using (a) and (b), show that $\Delta u = u_{xx} + u_{yy} = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\varphi\varphi}$.

Solution.

Problem 4.1

Let $\Omega \subseteq \mathbb{C}$ be convex and $f \in \mathcal{H}(\Omega)$. Show that if $\operatorname{Re}(f') > 0$ on Ω , then f is injective.

Solution.

Problem 4.2

A simple special case of Fubini's theorem.

- (a) Let $[a, b], [c, d] \subseteq \mathbb{R}$ be compact intervals and $F : [a, b] \times [c, d] \rightarrow \mathbb{C}$ continuous. Show that $x \mapsto \int_c^d F(x, y) dy$ and $y \mapsto \int_a^b F(x, y) dx$ are continuous on $[a, b]$ and $[c, d]$ respectively.
- (b) Show that

$$\int_a^b \left(\int_c^d F(x, y) dy \right) dx = \int_c^d \left(\int_a^b F(x, y) dx \right) dy.$$

Hint: the integrals exist by (a). The identity holds for constant F ; derive the general case by subdividing the rectangle into small rectangles and approximating F suitably.

Solution. (a) The argument is symmetric for both x and y . It suffices to show one. Note the following bounds,

$$\begin{aligned} \left| \int_a^b F(x, y) \, dx - \int_a^b F(x, y_0) \, dx \right| &= \left| \int_a^b (F(x, y) - F(x, y_0)) \, dx \right| \\ &\leq \int_a^b |F(x, y) - F(x, y_0)| \, dx. \end{aligned}$$

This last thing would be an issue but we know that F is a continuous function over a compact set and thus is uniformly continuous. Thus $\forall \varepsilon > 0$ there exists a $\delta > 0$ such that if two points are within delta of each other $|F(\vec{x}) - F(\vec{y})| < \frac{\varepsilon}{b-a}$. Thus for all $|y - y_0| < \delta$ we have that

$$\left| \int_a^b F(x, y) \, dx - \int_a^b F(x, y_0) \, dx \right| < \varepsilon$$

thus we have uniform cont.

Problem 4.3

Let $\alpha, \beta : [0, 1] \rightarrow \mathbb{C}$ be piecewise smooth paths and $G : \alpha^* \times \beta^* \rightarrow \mathbb{C}$ continuous. Show that

$$\int_{\alpha} \left(\int_{\beta} G(z, w) \, dw \right) dz = \int_{\beta} \left(\int_{\alpha} G(z, w) \, dz \right) dw.$$

Hint: use Problem 4.2 to show that these integrals exist and are equal.

Solution.